

Assignment #10: dp & bfs

Updated 2 GMT+8 Nov 25, 2024

2024 fall, Compiled by 吴诚舟---物理学院

说明:

- 1) 请把每个题目解题思路（可选），源码Python, 或者C++（已经在Codeforces/Openjudge上AC），截图（包含Accepted），填写到下面作业模版中（推荐使用 typora <https://typoraio.cn>，或者用 word）。AC 或者没有AC，都请标上每个题目大致花费时间。
- 2) 提交时候先提交pdf文件，再把md或者doc文件上传到右侧“作业评论”。Canvas需要有同学清晰头像、提交文件有pdf、“作业评论”区有上传的md或者doc附件。
- 3) 如果不能在截止前提交作业，请写明原因。

1. 题目

LuoguP1255 数楼梯

dp, bfs, <https://www.luogu.com.cn/problem/P1255>

代码:

```
n = int(input())
dp = [1]*(n+1)
for i in range(2, n+1):
    dp[i] = dp[i-1]+dp[i-2]
print(dp[-1])
```

代码运行截图 (至少包含有"Accepted")

测试点信息	源代码																					
<table border="1"><thead><tr><th>#1</th><th>#2</th><th>#3</th><th>#4</th><th>#5</th><th>#6</th><th>#7</th></tr></thead><tbody><tr><td>AC</td><td>AC</td><td>AC</td><td>AC</td><td>AC</td><td>AC</td><td>AC</td></tr><tr><td>16ms/3.66MB</td><td>16ms/3.68MB</td><td>16ms/3.72MB</td><td>16ms/3.77MB</td><td>15ms/3.68MB</td><td>16ms/3.68MB</td><td>16ms/3.67MB</td></tr></tbody></table>	#1	#2	#3	#4	#5	#6	#7	AC	AC	AC	AC	AC	AC	AC	16ms/3.66MB	16ms/3.68MB	16ms/3.72MB	16ms/3.77MB	15ms/3.68MB	16ms/3.68MB	16ms/3.67MB	
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<table border="1"><thead><tr><th>#8</th><th>#9</th><th>#10</th></tr></thead><tbody><tr><td>AC</td><td>AC</td><td>AC</td></tr><tr><td>16ms/3.75MB</td><td>16ms/3.96MB</td><td>17ms/4.96MB</td></tr></tbody></table>	#8	#9	#10	AC	AC	AC	16ms/3.75MB	16ms/3.96MB	17ms/4.96MB													
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Python 3 101B 160ms 4.96MB

ruoGwcZ

所属题目 P1255 数楼梯

评测状态 Accepted

评测分数 100

提交时间 2024-11-26 16:04:18

27528: 跳台阶

dp, <http://cs101.openjudge.cn/practice/27528/>

代码:

```
n = int(input())
dp = [1]*(n+1)
for i in range(2, n+1):
    dp[i] = sum(dp[:i])
print(dp[-1])
```

代码运行截图 (至少包含有"Accepted")

#47408393提交状态

查看 提交 统计 提问

状态: Accepted

源代码

```
n = int(input())
dp = [1]*(n+1)
for i in range(2, n+1):
    dp[i] = sum(dp[:i])
print(dp[-1])
```

基本信息

#: 47408393
题目: 27528
提交人: 24n2400011447
内存: 3624kB
时间: 31ms
语言: Python3
提交时间: 2024-11-26 16:07:37

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English 帮助 关于

474D. Flowers

dp, <https://codeforces.com/problemset/problem/474/D>

代码:

```
t, k = map(int, input().split())

MOD = 1000000007
MAX = 100000
answer = []

dp = [1]*(MAX+1)
for i in range(2, MAX+1):
    if k == 1:
        dp[1] = 2
    if i-k >= 0:
        dp[i] = (dp[i-1]+dp[i-k])%MOD
    else:
        dp[i] = dp[i-1]

s = 0
pre_sum = [0]
for i in range(1, len(dp)):
    s += dp[i]
    pre_sum.append(s)
for i in range(t):
    a, b = map(int, input().split())
```

```

        answer.append((pre_sum[b]-pre_sum[a-1])%MOD)
    for ans in answer:
        print(ans)

```

代码运行截图 (至少包含有"Accepted")

The screenshot shows a LeetCode submission interface. At the top, there's a navigation bar with tabs: PROBLEMS, SUBMIT CODE, MY SUBMISSIONS, STATUS, HACKS, ROOM, STANDINGS, and CUSTOM INVOCATION. Below this is a 'General' tab. A table displays submission details for problem 474D - 15:

#	Author	Problem	Lang	Verdict	Time	Memory	Sent	Judged	
293328543	Practice: czwuuu	474D - 15	Python 3	Accepted	515 ms	14696 KB	2024-11-26 11:50:15	2024-11-26 11:50:15	★ Compare

Below the table is a 'Source' tab showing the code. The code is a Python script that reads input, calculates a dynamic programming array 'dp', and prints the results. The code is as follows:

```

t, k = map(int, input().split())
MOD = 1000000007
MAX = 100000
answer = []

dp = [1]*(MAX+1)
for i in range(2, MAX+1):
    if k == 1:
        dp[i] = 2
    if i-k >= 0:
        dp[i] = (dp[i-1]+dp[i-k])%MOD
    else:
        dp[i] = dp[i-1]

s = 0
pre_sum = [0]
for i in range(1, len(dp)):
    s += dp[i]
    pre_sum.append(s)
for i in range(t):
    a, b = map(int, input().split())
    answer.append((pre_sum[b]-pre_sum[a-1])%MOD)
for ans in answer:
    print(ans)

```

LeetCode5.最长回文子串

dp, two pointers, string, <https://leetcode.cn/problems/longest-palindromic-substring/>

代码:

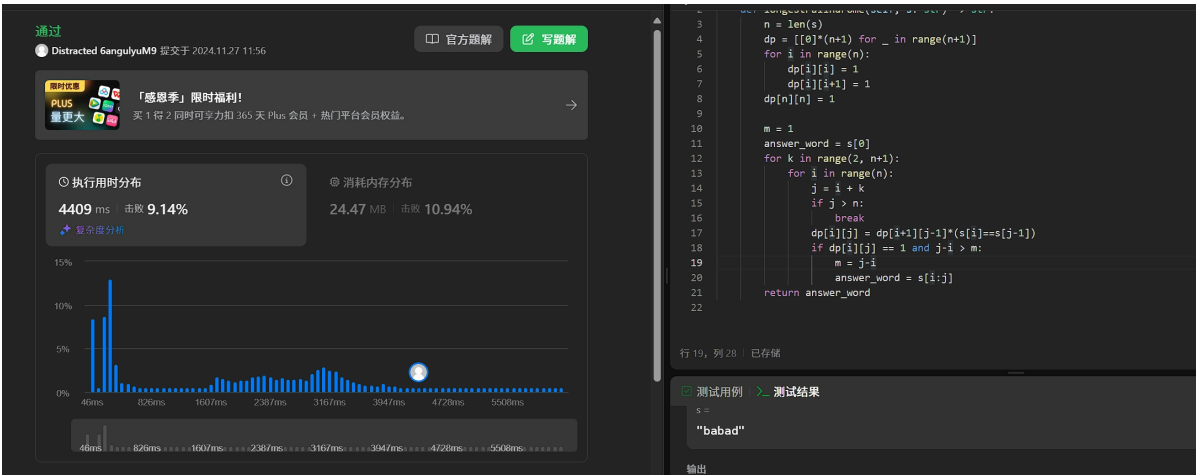
```

class Solution:
    def longestPalindrome(self, s: str) -> str:
        n = len(s)
        dp = [[0]*(n+1) for _ in range(n+1)]
        for i in range(n):
            dp[i][i] = 1
            dp[i][i+1] = 1
        dp[n][n] = 1

        m = 1
        answer_word = s[0]
        for k in range(2, n+1):
            for i in range(n):
                j = i + k
                if j > n:
                    break
                dp[i][j] = dp[i+1][j-1]*(s[i]==s[j-1])
                if dp[i][j] == 1 and j-i > m:
                    m = j-i
                    answer_word = s[i:j]
        return answer_word

```

代码运行截图 (至少包含有"Accepted")



12029: 水淹七军

bfs, dfs, <http://cs101.openjudge.cn/practice/12029/>

搞了2h发现是输入问题...

代码:

```
from collections import deque

direction = [(1, 0), (0, 1), (-1, 0), (0, -1)]
def bfs(xw, yw, field, I, J):
    queue = deque(); queue.append((xw, yw))
    if xw == I and yw == J:
        return True

    while queue:
        x, y = queue.popleft()
        for i in range(4):
            nx, ny = x + direction[i][0], y + direction[i][1]
            if field[nx][ny] < field[x][y]:
                field[nx][ny] = field[x][y]
                if nx == I and ny == J:
                    return True
                queue.append((nx, ny))
    return False

k = int(input())
answer = []
for _ in range(k):
    #input
    M, N = map(int, input().split())
    field = [[99999]*(N+2)]+[[99999]+list(map(int, input().split()))+[99999] for
_ in range(M)]+[[99999]*(N+2)]
    I, J = map(int, input().split())
    P = int(input())
```

```

ans = False

#process
for _ in range(P):
    xw, yw = map(int, input().split())
    ans = bfs(xw, yw, field, I, J)
    if ans:
        break
answer.append(ans)

for a in answer:
    print('Yes' if a else 'No')

```

代码运行截图 (至少包含有"Accepted")

#47413709提交状态

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[统计](#)
[提问](#)

状态: Runtime Error

源代码

```

from collections import deque

direction = [(1, 0), (0, 1), (-1, 0), (0, -1)]
def bfs(xw, yw, field, I, J):
    queue = deque([(xw, yw)])
    if xw == I and yw == J:
        return True

    while queue:
        x, y = queue.popleft()
        for i in range(4):
            nx, ny = x + direction[i][0], y + direction[i][1]
            if field[nx][ny] < field[x][y]:
                field[nx][ny] = field[x][y]
                if nx == I and ny == J:
                    return True
            queue.append((nx, ny))
    return False

k = int(input())
answer = []
for _ in range(k):
    #input
    M, N = map(int, input().split())
    field = [[9999]*(N+2)] + [[9999]+list(map(int, input().split()))+[9999]]*(M-1)
    I, J = map(int, input().split())
    P = int(input())
    ans = False

```

基本信息

: 47413709
题目: 12029
提交人: 24n2400011447
内存: 3616kB
时间: 30ms
语言: Python3
提交时间: 2024-11-26 20:01:13

不想改输入输出了 (怒, 我确定自己是对的)

02802: 小游戏

bfs, <http://cs101.openjudge.cn/practice/02802/>

代码:

```

from collections import deque

direction = [[1, 0], [0, -1], [-1, 0], [0, 1]]
def connect(mat, w, h, x1, y1, x2, y2):
    queue = deque(); queue.append((x1, y1, -1, 0))
    inq = set(); inq.add((x1, y1))
    recorder = []
    if x1 == x2 and y1 == y2:
        return 0

```

```

while queue:
    x, y, dire, turn = queue.popleft()
    for i in range(4):
        nx, ny = x+direction[i][0], y+direction[i][1]
        if (nx, ny) in inq or nx < 0 or nx > w+1 or ny < 0 or ny > h+1:
            continue
        if mat[ny][nx] != 'X' or (nx == x2 and ny == y2):
            turn_new = turn+1 if dire != i else turn
            if nx == x2 and ny == y2:
                recorder.append(turn_new)
            else:
                inq.add((nx, ny))
                queue.append((nx, ny, i, turn_new))
    return min(recorder) if recorder else -1

#输入和答案的存储
n = 0
answer = []
while True:
    n += 1
    w, h = map(int, input().split())
    mat = [" "]*(w+2)
    if w == 0 and h == 0:
        break
    else:
        for i in range(h):
            line = [" "]+list(input())+" "
            mat.append(line)
        mat.append([" "]*(w+2))

    m = 0
    _answer_ = []
    while True:
        m += 1
        x1, y1, x2, y2 = map(int, input().split())
        if x1 == 0 and x2 == 0 and y1 == 0 and y2 == 0:
            break
        else:
            _answer_.append(connect(mat, w, h, x1, y1, x2, y2))
    answer.append(_answer_)

#输出模块
for i in range(1, n):
    print(f"Board #{i}:")
    for j in range(1, len(answer[i-1])+1):
        if answer[i-1][j-1] == -1:
            ans_word = "impossible."
        else:
            ans_word = str(answer[i-1][j-1])+" segments."
        print(f"Pair {j}: {ans_word}")
    print('')

```

状态: Accepted

源代码

```
from collections import deque

direction = [[1, 0], [0, -1], [-1, 0], [0, 1]]
def connect(mat, w, h, x1, y1, x2, y2):
    queue = deque(); queue.append((x1, y1, -1, 0))
    inq = set(); inq.add((x1, y1))
    recorder = []
    if x1 == x2 and y1 == y2:
        return 0

    while queue:
        x, y, dire, turn = queue.popleft()
        for i in range(4):
            nx, ny = x+direction[i][0], y+direction[i][1]
            if (nx, ny) in inq or nx < 0 or nx > w+1 or ny < 0 or ny > h+1:
                continue
            if mat[ny][nx] != 'X' or (nx == x2 and ny == y2):
                turn_new = turn+1 if dire != i else turn
                if nx == x2 and ny == y2:
                    recorder.append(turn_new)
                else:
                    inq.add((nx, ny))
                    queue.append((nx, ny, i, turn_new))
    return min(recorder) if recorder else -1
```

基本信息

#: 47420600
题目: 02802
提交人: 24n2400011447
内存: 3920kB
时间: 48ms
语言: Python3
提交时间: 2024-11-27 11:19:56

2. 学习总结和收获

如果作业题目简单，有否额外练习题目，比如：OJ“计概2024fall每日选做”、CF、LeetCode、洛谷等网站题目。

前两道题十分简单。最后两道bfs花了很长时间调试，其中水淹七军居然是测试数据有问题，破大防。

最近每日选做有点落下了。